$\mathbb{R}$.

Prop. $(0,1) \cong \mathbb{R}$.

Proof. Define $f: (0,1) \to \mathbb{R} : x \mapsto \log \frac{x}{1-x}$.

then, $f^{-1}: \mathbb{R} \to (0,1) : x \mapsto \frac{e^x}{1+e^x}$

is the inverse, and $f, f^{-1}$ both are continuous, so proved. ☑

Prop. $[0,1] \cong [a,b]$ for all $a<b$.

Proof. Define a map

$$L: [0,1] \to [a,b] : t \mapsto (1-t)\cdot a + t \cdot b.$$

with the inverse is

$$L^{-1}: [a,b] \to [0,1] : t \mapsto \frac{1}{b-a}(t-a)$$